

CLARUS Analysis Report

Document: rfc793.txt
Analyzed: 2026-05-06T14:50:32.963454

1. Document Processing

Metric	Value
Total Segments	2106
Normative Segments	364
Informative Segments	808
Unknown Segments	934
High Confidence (≥ 0.7)	383
Low Confidence (< 0.4)	0
Text Length (chars)	177957

2. Semantic Extraction

Metric	Value
Total Profiles	30
Average Confidence	70.6%

Profiles (30)

#	Sentence	Subject	Object	Modality	Condition
1	A gateway is permitted to break up an internet datagram into smaller segments, if necessary for transmission through the next network	gateway	reassemble fragments into original datagram	is permitted to	
2	send that data in segments at its own convenience, until the push flag is set	the push flag	send all unsent data	must	a
3	There is no guarantee that senders will use this option, so they must not begin on a word boundary	they	use this option	must	they do not begin on a word boundary
4	Note that when the receive window is zero no segments should be sent	no segments	be	should	the receive window is zero no segments should be sent
5	ACKs. However, even when the receive window is zero, a receiver must not send an ACK until the receive window is greater than zero	ACKs	window is zero	must	the receive window is zero
6	(MSL) before assigning any sequence numbers upon starting up or restarting	starting up or restarting	was lost	may	experience indicates it is desirable to do so
7	lifetime in the net is not likely to exceed a few tens of seconds, and the amount of data in transit is not likely to exceed a few tens of megabits	lifetime	a little short	may	data rates escalate to 10 ⁷ of megabits/sec
8	The basic duplicate detection and sequencing algorithm in TCP can be defeated	init TCP can be	defeated	can	a source TCP does not have any memory of the sequence numbers it last used on a given connection

9	incoming segment) the precedence level of the incoming segment must...	sent	must	it does not a reset must be sent	
10	FIN segments to be exchanged. When all segments preceding the FINs...	ACK the FIN it	can	all segments preceding the FINs have been processed and acknowledged	
11	must tell the user to go into "urgent mode"; when the...	TCP receive...	tell user to go	must	the receive sequence number catches up to the urgent pointer
12	If the sending TCP simply sends as much as it can whenever the	whenever the	can	the sending TCP simply	sends as much as it c...
13	If the active/passive flag is set to passive, then this is a call t...	passive open	to passive	may	the active/passive flag is set to passive
14	This call causes the data contained in the indicated user buffer to...	Some implementations	allow users to SEND data	may	the connection has not been opened
15	If the PUSH flag is set, the data must be transmitted promptly to t...	PUSH flag	transmitted promptly to the receiver	must	the PUSH flag is set
16	may be trying to transmit the last of its data. Thus, CLOSE means ...		receive any more	may	the user level protocol is not well thought out) that the closing side is un... of all its data before timing out
17	Depending on the state of the connection, or on the implementation ...	some of this information	available or meaningful	may not	the calling process is not authorized to use this connection
18	urgent bit if requested in the command must be sent with the data	the data	sent with the data	must	requested in the command must be sent with t...
19	Strictly speaking, this is an error and should receive a "retransmission"	retransmission	acceptable	should	a second FIN is not emitted (the first
20	Any acknowledgment is bad if it arrives on a connection still in the...	still in	be	should	it arrives on a connection still in the LISTEN state
21	SEG.SEQ. SND.UNA should be advanced to equal SEG.ACK (if this is the case) and the...	ACK (if this is the case)	advanced to equal SEG.ACK	should	there is an ACK)
22	transmission may be included. If there are other controls or text ...	transmission	included	may	there are other controls or text in the segment then continue processing a...
23	Segments are processed in sequence. Initial tests on arrival are u...	new parts	be	should	a segment's contents straddle the boundary between old and new
24	SYN and FIN), and only processing further if the segment begins...	beginning sequence numbers	held for later processing	may	the segment then begins at RCV
25	LISTEN state and return. The user need not be informed. If this c...		removed	should	this connection was initiated with an active OP...
26	If the RST bit is set then, any outstanding RECEIVES and SENDs should...	SENDs	flushed	should	the RST bit is set then
27	If the SYN is in the window it is an error, send a reset, any it outst...		flushed	should	the SYN is in the window it is an error
28	(SEG.ACK & SND.UNA), it can be ignored. If the ACK acknowledges something...	acks	ignored	can	the ACK acks something not yet sent (SEG
29	If SND.UNA & SEG.ACK = SND.NXT, the send window should be updated...	send window	updated	should	SND
30	Once in the ESTABLISHED state, it is possible to deliver segment te...	segment	buffers	can	the segment empties and carries an PUSH flag

3. Terminology Analysis

Metric	Value
Defined Terms	0
Undefined Terms	50
Definition Coverage	0.0%

4. Contradiction Analysis

Total contradictions detected: **65**

#	Type	Conf.	Statement A	Statement B
1	semantic	100%	Otherwise return "error: connection does not e...	Return "state = ESTABLISHED", and the TCB ...
2	semantic	100%	Return "state = SYN-RECEIVED", and the TCB...	Return "state = CLOSE-WAIT", and the TCB p...
3	semantic	100%	(which is not always possible), and so it must ask t...	not wait at all; it must only be sure to use sequenc...
4	semantic	100%	Return "state = SYN-SENT", and the TCB poi...	Return "state = FIN-WAIT-1", and the TCB p...
5	semantic	100%	Return "state = SYN-SENT", and the TCB poi...	Return "state = FIN-WAIT-2", and the TCB p...
6	semantic	100%	Otherwise return "error: connection does not e...	Return "state = SYN-RECEIVED", and the TCB...
7	semantic	100%	Return "state = SYN-RECEIVED", and the TCB...	Return "state = FIN-WAIT-2", and the TCB p...
8	semantic	100%	Return "state = SYN-RECEIVED", and the TCB...	Return "state = CLOSING", and the TCB poin...
9	semantic	100%	Return "state = SYN-RECEIVED", and the TCB...	Return "state = FIN-WAIT-1", and the TCB p...
10	semantic	100%	If the foreign socket is specified, then change the ...	Foreign socket was not specified, then return "...
11	semantic	100%	Do the same processing as for the ESTABLISHED state.	delete the TCB, enter the CLOSED state, and return.
12	semantic	100%	the network, the TCP must keep quiet for a maximum s...	not wait at all; it must only be sure to use sequenc...
13	semantic	100%	exactly match the security/compartment and precedenc...	"connection reset" signal. Enter the CLOS...
14	semantic	100%	RECEIVE until signaled that a RECEIVE failed because...	can terminate his side gracefully. A TCP will relia...
15	semantic	100%	Return "state = SYN-SENT", and the TCB poi...	Return "state = CLOSE-WAIT", and the TCB p...
16	semantic	100%	Otherwise return "error: connection does not e...	Return "state = LISTEN", and the TCB pointer.
17	semantic	100%	Return "state = CLOSING", and the TCB poin...	Return "state = TIME-WAIT", and the TCB po...
18	semantic	100%	Return "state = FIN-WAIT-1", and the TCB p...	Return "state = CLOSE-WAIT", and the TCB p...
19	semantic	100%	Return "state = FIN-WAIT-2", and the TCB p...	Return "state = CLOSE-WAIT", and the TCB p...
20	semantic	100%	segments must carry current acknowledgment information...	covers that sequence number (segments sent do not ha...
21	semantic	100%	Otherwise return "error: connection does not e...	Return "state = SYN-SENT", and the TCB poi...
22	semantic	100%	(which is not always possible), and so it must ask t...	the network, the TCP must keep quiet for a maximum s...
23	semantic	100%	The precedence in the segment must match the precede...	If a reset was sent, discard the segment and return.
24	semantic	100%	Return "state = ESTABLISHED", and the TCB...	Return "state = FIN-WAIT-1", and the TCB p...
25	semantic	100%	arrives than can be accepted, it will be discarded. ...	When the receiving TCP has a zero window and a segme...
26	semantic	100%	Return "state = LISTEN", and the TCB poin...	Return "state = ESTABLISHED", and the TCB ...
27	semantic	100%	Return "state = ESTABLISHED", and the TCB...	Return "state = FIN-WAIT-2", and the TCB p...

28	semantic	100%	Return "state = CLOSE-WAIT";, and the TCB p...	Return "state = TIME-WAIT";, and the TCB po...
29	semantic	100%	Return "state = ESTABLISHED";, and the TCB p...	Return "state = CLOSING";, and the TCB poi...
30	semantic	100%	Return "state = LISTEN";, and the TCB pointer	Return "state = SYN-SENT";, and the TCB poi...
31	semantic	100%	urgent bit if requested in the command must be sent ...	queue the request, respond with "error: insuff...
32	semantic	100%	For a connection to be established or initialized, t...	The synchronization requires each side to send it&#x...
33	semantic	100%	incoming segment) the precedence level of the incomi...	unacceptable acknowledgment number) must elicit only...
34	semantic	100%	If the caller does not have access to the local sock...	If Foreign socket was not specified, then return &qu...
35	semantic	100%	urgent data, but the general notion is that the rece...	TCP implementations will follow a general principle ...
36	semantic	100%	Otherwise return "error: connection does not e...	Return "state = FIN-WAIT-1";, and the TCB p...
37	semantic	100%	Return "state = SYN-RECEIVED";, and the TCB p...	Return "state = ESTABLISHED";, and the TCB ...
38	semantic	100%	the default values must be used.	allowed to specify that the precedence must be exact...
39	semantic	100%	covers that sequence number (segments sent do not ha...	meaningful and must be added to the segment sequence...
40	semantic	100%	Remain in the TIME-WAIT state. Restart the 2 MSL ti...	For any state if the user timeout expires, flush all...
41	semantic	99%	independently by each TCP they might not be unique. ...	will be unique throughout all networks connected tog...
42	semantic	99%	RECEIVE until signaled that a RECEIVE failed because...	data in return need only wait to hear the connection...
43	semantic	99%	To avoid confusion we must prevent segments from one...	Since we assume that segments will stay in the netwo...
44	semantic	99%	Return "state = LISTEN";, and the TCB pointer	Return "state = SYN-RECEIVED";, and the TCB...
45	semantic	99%	In a connection with a one-way data flow, the window...	window is non zero, the transmitted data will be bro...
46	semantic	99%	Return "state = ESTABLISHED";, and the TCB p...	Return "state = CLOSE-WAIT";, and the TCB p...
47	semantic	99%	To distinguish among several outstanding RECEIVES an...	the sense that outstanding SENDs will be transmitted...
48	semantic	99%	match the security/compartiment in the TCB, send a rese...	If a reset was sent, discard the segment and return.
49	semantic	99%	below where the URG bit is checked, otherwise return.	Otherwise enter SYN-RECEIVED, form a SYN,ACK segment
50	semantic	99%	to stimulate the receiving user to accept some urgen...	is in "urgent mode";, the update will be in...
51	semantic	99%	If the caller does not have access to the local sock...	queue the request, respond with "error: insuff...
52	semantic	98%	On an active OPEN command, the TCP will begin the pr...	will abort the connection. The present global defau...
53	semantic	98%	Return "state = FIN-WAIT-1";, and the TCB p...	Return "state = FIN-WAIT-2";, and the TCB p...
54	semantic	98%	unacceptable acknowledgment number) must elicit only...	TCP. Users must keep reading connections they close...
55	semantic	97%	sent has been acknowledged then the three variables ...	covers that sequence number (segments sent do not ha...
56	semantic	97%	To avoid confusion we must prevent segments from one...	from the other side. Each side must also receive th...
57	semantic	97%	If active and the foreign socket is specified, then ...	If Foreign socket was not specified, then return &qu...
58	semantic	96%	must not be sent if it is not clear that this is the...	continue as if the precedence matched exactly (if th...
59	modality	90%	As a general rule, reset (RST) must be sent whenever...	must not be sent if it is not clear that this is the...
60	modality	90%	must not be sent if it is not clear that this is the...	incoming segment) the precedence level of the incomi...
61	modality	90%	must not be sent if it is not clear that this is the...	unacceptable acknowledgment number) must elicit only...
62	semantic	90%	acceptable except ACK segments. Thus, it is be poss...	control will be acted upon). Control information is...
63	semantic	87%	from the other side. Each side must also receive th...	not wait at all; it must only be sure to use sequenc...
64	semantic	82%	If the caller does not have access to the local sock...	If active and the foreign socket is specified, then ...

65	semantic	82%	re-open the connection. TCP B, in the meantime, thi...	detected a half-open connection. TCP B aborts at li...
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